Introduction

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0.1 Introduction

This project investigates different approaches to classification of mortality in the UCI Myocardial Infarction dataset (from [1]) based on patient symptoms after 24 hours, 48 hours, and 72hours, their medical history, age, and sex, blood chemistries, and other features. We have chosen to investigate logistic regression, decision trees, and random forests. We have spent a section for each looking at how to implement them on the data, before comparing their performances based on different metrics in our conclusion.

We began by preprocessing the data, which we started by removing all the columns which had more than 50% missing values, because we couldn't get meaningful insights from them within the scope of the report, and splitting the data into training, validation and testing data to be used consistently across sections so that we could compare performance. The data contains a lot of missingness, so we used simple imputation, which we chose to do individually to allow imputation to be done in each fold for k-fold cross validation. We also found that the data was highly imbalanced towards surviving patients, which was a particular problem because in this context we wanted to avoid being overly optimistic towards medical risk. Each section addresses the large false negative rate differently as appropriate to the model, and across sections we prioritised performance metrics which minimised this in some way.

The logistic regression section looks at feature selection, regularization, and splines regression, and comapares the performance of each, before looking at threshold adjusting to address the false negative rate. The decision tree section takes a different approach, first balacing the data and then creating a decision tree based on balanced data. The random forest section uses different performance metrics to tune random forests to the data using k-fold cross validation. The random forests section produced the most varied models with the widest range of performance metrics.

Throughout, we let TN, TP, FP, FN represent true negetaives, true positives, false negatives and false positives. We compare our models on 7 metrics: accuracy, precision, recall, f1, matthews correlation coefficient, balanced accuracy and receiver operating characteristic area under curve, which are explained below.

Accuracy is the proportion of correct predictions, Accuracy $= \frac{TP+TN}{TP+TN+FP+FN}$. As our data is quite unbalanced towards negatives, and we felt that our models should be more sensitive to positive cases given the medical comtext, so this metric was generally not prioritised in the report.

Precision is the proportion of predicted positives that are correct, Precision $= \frac{TP}{TP+FP}$. Again, we did not prioritise this metric as we found that due to the high number of negatives, any model that correctly identified a lot of the true positives ended up giving us a lot of false positives, making our precision quite low. Recall is the proportion of actual positives detected, Recall $= \frac{TP}{TP+FN}$. We heavily prioritised this throughout as we felt that the consequences for having a false negative

when assessing the risk of mortality for a patient was the worst outcome, and that unnecessary interventions caused by false positives were not as important to avoid.

The F1 score is their harmonic mean, $F_1 = \frac{2TP}{2TP+FP+FN}$. We use this correlation as a guardrail to ensure our model was still accounting for precision, but we did not prioritise it. The Matthews correlation coefficient, $\text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}}$, measures correlation between predicted and actual labels. Again, we used this as a sanity check to ensure we still had a good correlation between labels while trying to avoid false negatives, and we used it to see how much predicition power we were losing when adapting models to achieve higher recall.

Balanced accuracy averages sensitivity and specificity, $\text{BA} = \frac{1}{2}\left(\frac{TP}{TP+FN} + \frac{TN}{TN+FP}\right)$, giving each class equal weight. We used this instead of accuracy to give us a general idea of how well the model was doing without directly optimising over it.

ROC-AUC is the area under the true-positive-rate versus false-positive-rate curve across classification thresholds, $\text{AUC} = \int_0^1 \text{TPR}(u)\, du$, where $u = \text{FPR} = \frac{FP}{FP+TN}$ and $\text{TPR} = \frac{TP}{TP+FN}$. This is equivalent to the probability that the model assigns a higher score to a random positive result than a random negative result, which is proved in [1], and in section 2. This was very useful to compare models across different thresholds, particularly in section 1 and section {michal's section} where we used threshold adjustment to refine the model and achieve a better recall.

1. Golovenkin, S., Shulman, V., Rossiev, D., Shesternya, P., Nikulina, S., Orlova, Y., & Voino-Yasenetsky, V. (2020). Myocardial infarction complications [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C53P5M
2. Andreas Maurer, Massimiliano Pontil (2020), Estimating weighted areas under the ROC curve https://proceedings.neurips.cc/paper/2020/file/5781a2637b476d781eb3134581b32044-Paper.pdf
3. Singh M, Cariou A, Varenne O, Picard F, Pham V. Phenotypic clustering of myocardial infarction complicated by out-of-hospital cardiac arrest using unsupervised machine learning. Resuscitation. 2026 Jun;223:111037. doi: 10.1016/j.resuscitation.2026.111037. Epub 2026 Mar 5. PMID: 41794115.